

Project Proposal

Project Title

Design and Implementation of an Online Food Ordering System Using Spring Boot

Team Members

MD ANISUR RAHMAN, 2022331036 – Frontend development, admin panel UI implementation, API integration, and user experience testing

ABDULLAH OMAR SAYEEM, 2022331058 – Backend development, system architecture design, business logic implementation, and integration of design patterns

ANGKON ROY, 2022331060 – Database schema design, ER diagram preparation, transaction handling, and data consistency management

MUHAMMAD MAHIR ASEF, 2022331064 – Software Requirement Specification (SRS), use case and sequence diagram preparation, and requirement analysis

SYED RAFI ALAM, 2022331086 – Class diagram design, system testing, debugging, quality assurance, and project presentation preparation

Project Description

The Online Food Ordering System is a backend-centric web application that enables users to browse restaurant menus, place food orders, and track order status in real time. Restaurants and administrators can manage menus, process orders, and monitor transactions through a secure admin panel.

The system is developed using Java, Object-Oriented Programming principles, and Spring Boot, following standard software engineering practices such as modular design, layered architecture, and clear separation of concerns. Optional frontend components may be

implemented using HTML, CSS, Tailwind, and ReactJS, while the core emphasis remains on backend logic, transaction handling, and data consistency.

The project focuses on modeling real-world workflows such as order lifecycle management, payment processing, and error handling, making it suitable for demonstrating UML diagrams, SRS documentation, and design pattern usage.

Objectives

- To design a scalable and modular backend system using Spring Boot and OOP principles.
- To model real-world food ordering workflows, including menu browsing, cart management, checkout, and order tracking.
- To ensure data consistency and reliability during transactions such as order placement and payment processing.
- To apply standard software engineering practices, including requirement analysis, UML modeling, and layered architecture.
- To implement relevant design patterns such as Strategy, State, and Factory patterns.
- To develop a role-based admin panel for managing restaurants, menus, and orders.
- To enhance individual understanding of backend system design and reduce reliance on automated code generation tools.